E_1 PRIMACY IN MODULAR KOSZUL DUALITY: THE ORDERED BAR COMPLEX, THE AVERAGING MAP, AND FIVE INVARIANTS SURVIVING A Σ_n-QUOTIENT

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ABSTRACT. For a cyclic chiral algebra $\mathcal{A}$, the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct is the natural primitive object of the theory. The symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$, which underlies the classical Francis–Gaitsgory factorization picture, is its Σ_n-coinvariant image under an averaging map $\mathrm{av}\colon \mathfrak{g}^{E_1}_{\mathcal{A}} \twoheadrightarrow \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ of dg Lie algebras. The scalar modular characteristic $\kappa(\mathcal{A})$ is recovered at degree 2 from a meromorphic r-matrix $r(z)$ that lives natively on the ordered side: for abelian families the average is κ, while for non-abelian affine Kac–Moody it is the double-pole term κ_{dp}, and the full κ adds the Sugawara shift $\dim(\mathfrak{g})/2$. The Drinfeld associator is the degree-3 coinvariant of the KZ connection. The five foundational theorems A–D and H of modular Koszul duality admit ordered counterparts whose Σ_n-shadows are exactly the commutative theorems; information lost under av includes the full braiding, the Grothendieck–Teichmüller torsor of splittings, and the cross-degree leakage that obstructs a dg Lie section. Three distinct coalgebra structures feature and must not be conflated: the Harrison coLie structure with $n!/n$ terms at degree n, the coshuffle structure with 2^n terms, and the deconcatenation structure with $n+1$ terms. Only deconcatenation equips the ordered bar. We record three tiers of r-matrix input: the pole-free tier where $R = \tau$ is Koszul-signed, the vertex algebra tier where R is derived from local OPE, and the genuinely E_1 tier populated by Yangians and Etingof–Kazhdan quantum vertex algebras.

1. INTRODUCTION

The bar complex attached to an augmented associative algebra admits two very different presentations. In the first, one takes the reduced tensor coalgebra $T^c(s^{-1}\bar{A})$ with its deconcatenation coproduct; the bar differential is built from the associative product in consecutive slots. In the second, one passes to Σ_n-coinvariants and works with the symmetric bar complex, which is the natural home for commutative or E_∞-algebras and for the Francis–Gaitsgory chiral bar. The historical order in chiral algebra and factorization homology has been: symmetric bar first, ordered refinement second. This paper reverses the order of presentation, and with it the status of the two objects.

The thesis of this paper is that the ordered bar complex is the primitive object of modular Koszul duality, and that the symmetric bar is a derived shadow obtained by averaging. The averaging map is a surjective morphism of dg Lie algebras; all five foundational theorems of modular Koszul duality (Theorems A through D, together with the chiral Hochschild theorem H) lift to the ordered side, and each symmetric theorem is the Σ_n-coinvariant image of its ordered counterpart. The information lost under av is geometric, arithmetic, and physical: the spectral r-matrix, the KZ connection, the Drinfeld associator, and the full braiding of tensor categories of modules all live on the ordered side.

Three principles motivate the inversion. The first is operadic: Stasheff's associahedra K_n (cf. [10]) carry more information than the symmetric groups Σ_n, and the Getzler–Jones [6] theory of E_n-operads

Date: April 24, 2026.
2020 *Mathematics Subject Classification.* 17B69, 18M70, 18M85, 81T40.
Key words and phrases. Ordered bar complex, E_1-algebras, chiral algebras, Koszul duality, R-matrix, Drinfeld associator, modular operad, averaging, Grothendieck–Teichmüller group.

makes E_1 the lowest rung of a ladder whose every step is non-commutative. The second is representation-theoretic: the classical r-matrix of an affine Kac–Moody algebra at level k is the content of the degree-2 genus-0 Maurer–Cartan element on the ordered side, and its Σ_2-average recovers the degree-2 scalar shadow: for Heisenberg this is κ, while for non-abelian affine Kac–Moody it is $\kappa_{\mathrm{dp}} = k\dim(\mathfrak{g})/(2h^\vee)$, with the full κ obtained after adding the Sugawara shift $\dim(\mathfrak{g})/2$. The third is physical: the line-operator sector of a chiral conformal theory is naturally open and ordered, while the closed (bulk) sector is symmetric; the averaging map is the operadic image of the open-to-closed functor.

We work throughout with the conventions of the monograph [9], which we refer to as Volume I. Everything is stated with OPE modes rather than lambda-brackets; conformal weights are denoted h and desuspended degrees $|s^{-1}v| = |v| - 1$ (the bar complex lowers degree). The grading is cohomological.

1.1. **Notation and conventions.** Let k be a field of characteristic zero, X a smooth algebraic curve over k, and $\mathcal{A}$ an augmented cyclic chiral algebra on X in the sense of Beilinson–Drinfeld. We write $\bar{\mathcal{A}}$ for the augmentation ideal $\ker(\varepsilon\colon \mathcal{A} \to k)$. All bar constructions are defined on $\bar{\mathcal{A}}$, not on $\mathcal{A}$ itself. Symmetric groups Σ_n act on tensor powers in the standard way; coinvariants are left exact and are denoted with a lower subscript, $(-)_{\Sigma_n}$. The ordered and symmetric Fulton–MacPherson compactifications are written $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$ and $\overline{\mathrm{FM}}_n(\mathbb{C})$ respectively; the former is the Ran-space blow-up along ordered collision diagonals.

2. Three coalgebra structures

Before writing any bar complex we must distinguish three cooperadic structures carried by a sum of tensor powers. These are distinct both as vector spaces (through the number of summands in the coproduct) and as natural transformations. All three arise somewhere in modular Koszul duality; only one equips the ordered bar complex of this paper.

Definition 2.1 (The three coalgebra structures on $T(V)$)**.** Let V be a graded vector space and $T(V) = \bigoplus_{n\geq 0} V^{\otimes n}$ its tensor space. Three cooperad structures on $T(V)$ (or on the ordered/symmetric variants) arise naturally:

(i) The *Harrison coLie* (or co-Lie) cooperad Lie^c, whose degree-n component has dimension $(n-1)!$ and whose coproduct decomposes a Lie word into pairs of sub-Lie-words using the Harrison differential.

(ii) The *coshuffle cooperad* Sym^c, whose degree-n coproduct produces 2^n summands by the shuffle formula
$$\Delta_{\mathrm{sh}}(v_1\cdots v_n) = \sum_{I\sqcup J=\{1,\ldots,n\}} \varepsilon(I,J)\,(v_I)\otimes(v_J),$$
with the sum running over all 2^n bipartitions and $\varepsilon(I,J)$ the Koszul sign.

(iii) The *deconcatenation cooperad* T^c, whose degree-n coproduct produces $n+1$ summands by splitting the word at a single position:
$$\Delta(v_1\cdots v_n) = \sum_{i=0}^{n}(v_1\cdots v_i)\otimes(v_{i+1}\cdots v_n).$$

The three structures are *not* interconvertible without loss of information. At degree 3 the Harrison differential sees 2 terms, the coshuffle sees 8, and deconcatenation sees 4. Only the last is ordered and strictly associative at the coalgebra level.

Proposition 2.2 (Coshuffle is not deconcatenation). *Let V be nonzero. The coshuffle coproduct on $\mathrm{Sym}(V)$ and the deconcatenation coproduct on $T^c(V)$ are distinct cooperadic structures: they have different degree-n term counts (2^n vs. $n+1$), different symmetry properties (the coshuffle is cocommutative, deconcatenation is not), and different primitive subspaces.*

Proof. Cocommutativity of the coshuffle follows from the involution $I \leftrightarrow J$ on bipartitions. Deconcatenation does not admit such an involution: swapping $(v_1 \cdots v_i)$ with $(v_{i+1} \cdots v_n)$ does not preserve the ordered word. The term counts are elementary. □

The three cooperads sit in a commutative square of natural transformations whose top arrow is the symmetrization and whose bottom arrow is the Harrison projection, but they are not canonically isomorphic. Conflating coshuffle with deconcatenation is a standard error in the chiral-algebra literature and produces spurious identifications between factorization coproducts and bar coproducts.

Remark 2.3 (Where each structure appears). The Harrison coLie cooperad Lie^c underlies the chiral Lie bar complex used by Francis–Gaitsgory, where only the zeroth product of the OPE is visible. The coshuffle cooperad Sym^c underlies the symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ on the Ran space, where factorization is implemented by the topological coproduct on unordered configuration space. The deconcatenation cooperad T^c underlies the ordered bar $\overline{B}^{\mathrm{ord}}(\mathcal{A})$, where the coproduct splits a word along a single internal edge and remembers the ordering on the collision points. The averaging map av of Section 4 descends from T^c to Sym^c by external Σ_n-coinvariants; it does not factor through Harrison unless one further restricts to the Eulerian weight-1 summand.

3. The ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A})$

We now define the ordered bar complex of a cyclic chiral algebra. The construction is standard in the sense that every ingredient appears in the chiral-algebra literature, but the insistence on the deconcatenation coproduct at the level of the ordered configuration space makes the E_1 structure manifest.

Definition 3.1 (The ordered bar complex). Let $\mathcal{A}$ be a strongly admissible augmented E_1-chiral algebra on X, with augmentation ideal $\bar{\mathcal{A}} = \ker \varepsilon$. The *ordered bar complex* is the graded vector space

$$\overline{B}^{\mathrm{ord}}(\mathcal{A}) \;:=\; T^c\big(s^{-1}\bar{\mathcal{A}}\big) \;=\; \bigoplus_{n\geq 0}\big(s^{-1}\bar{\mathcal{A}}\big)^{\otimes n}, \tag{3.1}$$

equipped with the deconcatenation coproduct $\Delta([a_1|\cdots|a_n]) = \sum_{i=0}^{n}[a_1|\cdots|a_i]\otimes[a_{i+1}|\cdots|a_n]$ and the bar differential d_{bar} obtained by extracting OPE collision residues at consecutive pairs of points in the ordered configuration space. The augmentation ideal $\bar{\mathcal{A}}$ is essential here: using $\mathcal{A}$ in place of $\bar{\mathcal{A}}$ in (3.1) would include the unit and break the construction.

The desuspension s^{-1} lowers the degree by one, so $|s^{-1}a| = |a| - 1$. The bar differential lowers tensor degree by one; the coproduct is a chain map with respect to d_{bar} by associativity of the chiral product in consecutive slots. Both the differential and the coproduct are built from the *ordered* collision pattern on $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$; the form $\eta = d\log(z_i - z_j)$ is the universal Arnold propagator (weight one, not weight h).

Theorem 3.2 (Ordered bar differential squared). *The ordered bar differential satisfies $d_{\mathrm{bar}}^2 = 0$.*

Proof. The argument is a codimension-two face pairing on $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$ with the Arnold relation supplying the required cancellation between the two ways of contracting an adjacent pair of collisions. See [9, Chapter 3] for the full argument and the relation to the ambient Mok25 compactification. □

Remark 3.3 (Three bar complexes, one map). Let $\overline{B}^{\mathrm{ch}}_{\mathrm{FG}}(\mathcal{A})$ denote the Francis–Gaitsgory chiral Lie bar, which uses only the zeroth OPE product $a_{(0)}b$; let $\overline{B}^{\Sigma}(\mathcal{A}) = \overline{B}^{\mathrm{ch}}(\mathcal{A})$ denote the symmetric bar, which uses all OPE products and takes Σ_n-coinvariants (the bar complex of Volume I, Theorem A); and let $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ denote the ordered bar of (3.1). There is a natural surjection

$$\overline{B}^{\mathrm{ord}}(\mathcal{A}) \twoheadrightarrow \overline{B}^{\Sigma}(\mathcal{A}), \qquad \mathrm{gr}\,\overline{B}^{\Sigma}(\mathcal{A}) \twoheadrightarrow \overline{B}^{\mathrm{ch}}_{\mathrm{FG}}(\mathcal{A}),$$

where the first arrow is Σ_n-coinvariant projection and the second is associated graded along the OPE filtration. These three complexes must not be conflated: the Francis–Gaitsgory complex sees only the Lie part, the symmetric bar sees the full symmetric OPE, and the ordered bar sees the braiding as well.

Example 3.4 (Heisenberg ordered bar). Let $\mathcal{H}_k$ be the Heisenberg chiral algebra at level k, generated by a single field J of conformal weight $h = 1$. The desuspended generator has degree $|s^{-1}J| = 0$. At degree 2 the ordered bar component is

$$(s^{-1}\bar{\mathcal{H}}_k)^{\otimes 2} \cong k \cdot [J|J],$$

a one-dimensional space. The bar differential extracts the collision residue of $J(z_1)J(z_2) \sim k/(z_1 - z_2)^2$ against the Arnold form $\eta = d\log(z_1 - z_2)$, producing a single pole $k/(z_1 - z_2)$.

Remark 3.5 (Heisenberg R-matrix convention: curved vs flat). The Heisenberg algebra $\mathcal{H}_k$ admits two standard presentations of the ordered bar R-datum: the *flat* (classical, linear) form $r(z) = k/z$ used throughout this note, and the *curved* (quantum, exponentiated) form $R(z) = \exp(k\ \ /z)$ used in the main chapter and in five companion standalones. They are related by the exponential map: $R(z) = \exp(\ \ r(z))$, so that the classical r-matrix is the $\ \ ^{0}$ coefficient of $\log R$. At every order in the exponentiation satisfies the quantum Yang–Baxter equation because $r(z)$ is a Casimir (its two tensor slots commute); the curved form is the trivialization of the twist, the flat form its infinitesimal shadow. This is the curved $\leftrightarrow$ flat Heisenberg equivalence (Costello–Witten twist trivialization); at $= 0$ the classical limit recovers $r(z)$.

4. The averaging map av

The averaging map is the dg Lie morphism that realises the symmetric bar as the Σ_n-coinvariant shadow of the ordered bar. Its domain is the E_1 convolution dg Lie algebra, its codomain is the modular convolution dg Lie algebra of Volume I, and its kernel carries the information lost in the passage from the open to the closed sector.

Definition 4.1 (The E_1 modular convolution dg Lie algebra). Let $\mathcal{A}$ be a cyclic E_1-chiral algebra and let $\mathcal{M}_{\mathrm{Ass}}$ denote the ribbon modular operad with $\mathcal{M}_{\mathrm{Ass}}(g,n) = C_*(\overline{\mathcal{M}}^{\mathrm{rib}}_{g,n})$, the chain complex of the moduli of stable Riemann surfaces with n parametrised boundary circles. Define

$$\mathfrak{g}^{E_1}_{\mathcal{A}} := \prod_{\substack{g,n \\ 2g-2+n>0}} \mathrm{Hom}\big(\mathcal{M}_{\mathrm{Ass}}(g,n),\ \mathrm{End}_{\mathcal{A}}(n)\big).$$

The Hom is taken *without* Σ_n-equivariance: this is the structural feature distinguishing $\mathfrak{g}^{E_1}_{\mathcal{A}}$ from the modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ of Volume I, in which the Hom is Σ_n-equivariant and the source cooperad is Sym^c-type rather than T^c-type. The dg Lie structure on $\mathfrak{g}^{E_1}_{\mathcal{A}}$ is inherited from the Feynman transform $F\mathrm{Ass}$: the differential contracts internal edges of ribbon graphs, the bracket composes ribbon graphs.

Definition 4.2 (The averaging map)**.** With the notation of Definition 4.1, the *averaging map*

$$\mathrm{av}\,:\, \mathfrak{g}^{E_1}_{\mathcal{A}} \twoheadrightarrow \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}, \qquad \mathrm{av}(\phi)(g,n) \;:=\; \frac{1}{n!}\sum_{\sigma\in\Sigma_n} \sigma\cdot\big(\phi(g,n)\circ\iota^{\mathrm{rib}}_{g,n}\big),$$

is the Reynolds operator for the external Σ_n-action, applied after pullback along any choice of section $\iota^{\mathrm{rib}}_{g,n}\colon \mathcal{M}_{\mathrm{Com}}(g,n) \to \mathcal{M}_{\mathrm{Ass}}(g,n)$ of the ribbon-forgetting quotient. The image is Σ_n-invariant by construction, independent of the section up to boundary, and lies in the modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ of Volume I.

Theorem 4.3 (av is a surjective dg Lie morphism)**.** *The averaging map* $\mathrm{av}\colon \mathfrak{g}^{E_1}_{\mathcal{A}} \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ *is a surjective morphism of dg Lie algebras; equivalently, it commutes with both the differential (ribbon edge contraction descending to stable edge contraction) and the Lie bracket (ribbon graph composition descending to symmetric graph composition).*

Proof. Surjectivity follows from the fact that every Σ_n-invariant homomorphism from $\mathcal{M}_{\mathrm{Com}}(g,n)$ lifts, via any section of the ribbon-forgetting quotient, to a homomorphism from $\mathcal{M}_{\mathrm{Ass}}(g,n)$. Compatibility with the differential follows because the ribbon edge-contraction differential descends to the symmetric edge-contraction differential under external Σ_n-coinvariants; this is the Getzler–Kapranov comparison between ribbon and commutative Feynman transforms [7]. Compatibility with the Lie bracket follows because symmetrisation commutes with the convolution bracket on cooperadic Homs. $\square$

Theorem 4.4 (Degree-2 averaging: Heisenberg exact, affine KM with Sugawara shift)**.** *Let* $r(z) \in \mathrm{End}(V\otimes V)\otimes \mathcal{O}(*\Delta)$ *be the genus-0 degree-2 E_1 shadow of $\mathcal{A}$: the degree-2 component of the E_1 Maurer–Cartan element, in the pre-dualisation convention of Volume I (the shadow-archetype column of* [9, Example]*, equivalently the collision residue computed in $\mathcal{A}\otimes\mathcal{A}$ before applying the Koszul pairing). Then*

$$\mathrm{av}_{n=2}\big(r(z)\big) \;=\; \begin{cases} \kappa(\mathcal{A}), & \textit{for abelian Heisenberg families},\\[4pt] \kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k) := \dfrac{k\,\dim(\mathfrak{g})}{2h^\vee}, & \textit{for non-abelian affine Kac–Moody in the trace-form convention.}\end{cases}$$

In the affine Kac–Moody case the full modular characteristic is

$$\kappa(\widehat{\mathfrak{g}}_k) \;=\; \kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k) + \frac{\dim(\mathfrak{g})}{2} \;=\; \frac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee},$$

so the extra $\dim(\mathfrak{g})/2$ *is the Sugawara simple-pole shift. Concretely, for the principal examples:*

- *Heisenberg $\mathcal{H}_k$: $r(z) = k/z$, $\kappa(\mathcal{H}_k) = k$.*
- *Affine Kac–Moody $\widehat{\mathfrak{g}}_k$: $r(z) = k\Omega_{\mathfrak{g}}/z$ (the level k survives; at $k=0$ the r-matrix vanishes identically),* $\mathrm{av}(r(z)) = \kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k) = k\dim(\mathfrak{g})/(2h^\vee)$*, and* $\kappa(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$.
- *Virasoro* Vir_c*: $r(z) = (c/2)/z^3 + 2T/z$ (cubic plus simple pole, no quartic), and* $\kappa(\mathrm{Vir}_c) = c/2$.

Proof. The degree-2 Maurer–Cartan component is, by definition, the degree-2 E_1 shadow $r(z) \in \mathrm{End}(V^{\otimes 2})\otimes\mathcal{O}(*\Delta)$ in the pre-dualisation convention. Averaging over Σ_2 projects onto the trivial isotypic component. For abelian Heisenberg families this scalar coincides with the full modular characteristic. For non-abelian affine Kac–Moody in the trace-form convention $r(z) = k\Omega_{\mathfrak{g}}/z$, the averaging sees only the double-pole channel and yields $\kappa_{\mathrm{dp}} = k\dim(\mathfrak{g})/(2h^\vee)$; the simple-pole self-contraction contributes the additional Sugawara term $\dim(\mathfrak{g})/2$, so $\kappa = \kappa_{\mathrm{dp}} + \dim(\mathfrak{g})/2$. The example computations are recorded in [9, Chapter 5]; each formula is verified by the three independent paths of pole-order analysis, the limiting case at $k=0$ for the trace-form r-matrix, and cross-family consistency against the landscape census. $\square$

Theorem 4.5 (Degree-3 averaging: associator to cubic shadow). *Let $\Phi_{\mathrm{KZ}}(\mathcal{A}) \in \mathrm{End}(V^{\otimes 3})$ be the genus-0 degree-3 Maurer–Cartan component, which for affine Kac–Moody $\widehat{\mathfrak{g}}_k$ is the Drinfeld–Kontsevich KZ associator. Then*

$$\mathrm{av}_{n=3}\big(\Phi_{\mathrm{KZ}}(\mathcal{A})\big) \;=\; \mathfrak{C}(\mathcal{A}),$$

where $\mathfrak{C}(\mathcal{A})$ is the cubic shadow of Volume I, equivalently the Σ_3-coinvariant of the associator. The antisymmetric component $[\Omega_{12}, \Omega_{23}]$ lies entirely in $\ker(\mathrm{av})$: it is antisymmetric under the transposition $(1 \leftrightarrow 3)$, so its Σ_3-average vanishes. Only the symmetric products $\Omega_{12}\Omega_{23} + \Omega_{23}\Omega_{12}$ contribute to $\mathfrak{C}(\mathcal{A})$.

Proof. The Σ_3-isotypic decomposition of Φ_{KZ} in $\mathrm{End}(V^{\otimes 3})$ splits into the trivial, sign, and standard representations. The Reynolds operator $\mathrm{av} = (1/6)\sum_{\sigma\in\Sigma_3}\sigma$ projects onto the trivial isotypic component. A direct check on $[\Omega_{12}, \Omega_{23}]$ shows it is antisymmetric under $(1 \leftrightarrow 3)$ and therefore has trivial isotypic component zero. The symmetric products $\Omega_{ij}\Omega_{k\ell} + \Omega_{k\ell}\Omega_{ij}$ are the only surviving contributors. The identification of this coinvariant with the cubic shadow is Volume I Theorem D(iii). □

Proposition 4.6 (Non-splitting of av). *The short exact sequence of dg Lie algebras*

$$0 \longrightarrow \ker(\mathrm{av}) \longrightarrow \mathfrak{g}^{E_1}_{\mathcal{A}} \xrightarrow{\mathrm{av}} \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} \longrightarrow 0$$

does not split as dg Lie algebras. At each fixed degree n, the extension is Lie-trivial: the inclusion $\mathrm{End}^{\Sigma_n}(V^{\otimes n}) \hookrightarrow \mathrm{End}(V^{\otimes n})$ is a Lie subalgebra embedding, so the degree-n extension 2-cocycle vanishes. The obstruction is cross-degree: the bar differential reduces degree by one and is Σ_{n-2}-equivariant but not Σ_n-equivariant, so elements of $\ker(\mathrm{av}_n)$ can map under d_{bar} to elements with nonzero av_{n-1} component. The space of Lie sections of av compatible with the Maurer–Cartan equation is a torsor for the Grothendieck–Teichmüller pro-unipotent group GRT_1 of Drinfeld.

Proof. See [9, Chapter 19] for the detailed argument. The GRT_1-torsor structure follows from Drinfeld's theorem on the space of associators [2] and the Etingof–Kazhdan quantisation theorem [3]. □

5. The five theorems as averaging-invariants

The central structural result of modular Koszul duality is that the five theorems A, B, C, D, H admit ordered E_1 counterparts $\mathrm{A}^{E_1}, \ldots, \mathrm{H}^{E_1}$, and that each symmetric theorem is the image of its ordered counterpart under av. We state the five ordered theorems precisely, sketch the comparison, and highlight Theorem D as the cleanest example: the scalar modular characteristic κ is the degree-2 coinvariant of the spectral r-matrix $r(z)$.

Theorem 5.1 (Theorem A^{E_1}: ordered bar–cobar). *The genus-g ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A})(g,n)$ is an $F\mathrm{Ass}$-coalgebra with differential satisfying $d^2 = 0$, where $F\mathrm{Ass}$ is the Feynman transform of the ribbon modular operad. The bar and cobar functors form an adjunction*

$$(\Omega^{\mathrm{ch}})^{(g)} \dashv (\overline{B}^{\mathrm{ord}})^{(g)} : \mathsf{Alg}_{F\mathrm{Ass}} \rightleftarrows \mathsf{Coalg}_{F\mathrm{Ass}},$$

and coinvariant projection recovers Theorem A of Volume I.

Theorem 5.2 (Theorem B^{E_1}: ordered inversion on the Koszul locus). *Assume $\mathcal{A}$ lies on the E_1 Koszul locus, meaning the PBW-completeness hypothesis of* [9, Chapter 4] *holds for the ordered bar complex. Then at each genus g, the counit*

$$(\Omega^{\mathrm{ch}})^{(g)}\big(\overline{B}^{\mathrm{ord}}(\mathcal{A})(g,\bullet)\big) \;\xrightarrow{\sim}\; \mathcal{A}$$

is a quasi-isomorphism (the inversion $\Omega^{\mathrm{ch}} \circ \overline{B}^{\mathrm{ord}} \simeq \mathrm{id}$ on the Koszul locus, not a Koszul-duality statement), and the E_1 linear Koszul dual $\mathcal{A}^{!,E_1}$ obtained by linear duality on a finite-dimensional model of $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ is *an E_1-chiral algebra at each such genus. The scope of "all genera" here is contingent on the PBW-completeness hypothesis being propagated genus by genus via the MC perturbation lemma of* [9, Chapter 9].

Theorem 5.3 (Theorem C^{E_1}: ordered complementarity). *At each genus g and degree n with $2g-2+n>0$, the ordered complementarity takes the form*

$$Q_{g,n}^{E_1}(\mathcal{A}) + Q_{g,n}^{E_1}(\mathcal{A}^{!,E_1}) \;=\; H^*(\overline{\mathcal{M}}_{g,n}^{\mathrm{rib}}, Z^{E_1}(\mathcal{A})),$$

where $Z^{E_1}(\mathcal{A})$ is the E_1 centre, the space of R-matrix-equivariant elements. The scalar identity $\kappa + \kappa^! = 0$ for Kac–Moody and free-field families (and $\kappa + \kappa^! = 13$ for Virasoro) is the Σ_n-coinvariant image.

Theorem 5.4 (Theorem D^{E_1}: degree-2 package). *The formal ordered degree-2 shadow series*

$$R^{E_1,\mathrm{bin}}(z;\ \ ) \;=\; \sum_{g\geq 0} {}^{2g}\, r_g(z)$$

obtained from the genus-refined degree-2 projection of the E_1 Maurer–Cartan element is the E_1 degree-2 characteristic package of $\mathcal{A}$. It is universal, additive, antisymmetric under opposite-duality, and satisfies $\mathrm{av}(r_0(z)) = \kappa(\mathcal{A})$ *on abelian families, while for non-abelian affine Kac–Moody one has* $\mathrm{av}(r_0(z)) = \kappa_{\mathrm{dp}}(\mathcal{A})$ *and* $\kappa(\mathcal{A}) = \kappa_{\mathrm{dp}}(\mathcal{A}) + \dim(\mathfrak{g})/2$. *The coinvariant recovery of the scalar curvature from $r(z)$ is the content of Theorem 4.4. A full higher-genus line-side modular R-matrix interpretation requires additional Yangian input and is not proved in this paper.*

Theorem 5.5 (Theorem H^{E_1}: ordered chiral Hochschild). *For every genus g and complete $\mathcal{A}$-bimodule M, the genus-g ordered chiral Hochschild homology is computed coalgebraically by the genus-g ordered chiral coHochschild complex of the ordered bar coalgebra $C = \overline{B}^{\mathrm{ord}}(\mathcal{A})$, with twisted differential involving the curvature term $\kappa(\mathcal{A}) \cdot \lambda_g$ in the uniform-weight case. At genus $g \geq 2$ with multi-weight field content the scalar formula receives a cross-channel correction $\delta F_g^{\mathrm{cross}}$, so the statement above is strict only in the uniform-weight regime; the multi-weight version is proved in Volume I. The Σ_n-coinvariant recovers the symmetric chiral Hochschild complex of Theorem H.*

Remark 5.6 (Theorem D is the cleanest example). Among the five, Theorem D makes the E_1 primacy thesis most concrete. The scalar modular characteristic κ is a single number per family, and the family formulas are distinct: for W_N the harmonic number is $H_N = \sum_{j=1}^{N} 1/j$ and must not be copied from a different family. Its degree-2 lift is a meromorphic function of a single variable, the collision coordinate z. The statement $\mathrm{av}(r(z)) = \kappa$ says this exactly for Heisenberg, while for non-abelian affine Kac–Moody the same averaging recovers only κ_{dp} and the full κ adds the Sugawara shift $\dim(\mathfrak{g})/2$. The scalar shadow is still the Σ_2-average of a function. This is a strict projection; the full function $r(z)$ carries information about the braiding, the quantum group, and the line-operator algebra that no scalar invariant can see. The universality property of κ in Volume I Theorem D lifts (with the same proof, interpreted one level higher) to a universality property of $r(z)$ on the ordered side.

Remark 5.7 (What averaging forgets). The kernel of av contains several objects of independent interest. At degree 2, $\ker(\mathrm{av})$ is the antisymmetric part of $r(z)$; for bosonic generators of even desuspended degree this vanishes identically, and κ recovers all of $r(z)$. For generators of odd desuspended degree, the kernel is nontrivial and captures the parity-dependent Eulerian weight structure on the bar complex. At degree 3, $\ker(\mathrm{av})$ contains the linearised Drinfeld commutator $[\Omega_{12}, \Omega_{23}]$ and, by the non-splitting proposition, obstructs a canonical reconstruction of the full associator from $\mathfrak{C}$. The GRT_1-torsor structure of the splittings of av is the operadic shadow of Etingof–Kazhdan non-uniqueness of quantisation.

6. Three tiers of R-matrix

The ordered bar complex distinguishes three tiers of input, corresponding to how the degree-2 Maurer–Cartan component $r(z)$ is constructed. The tiers are a refinement of the E_n-hierarchy restricted to the degree-2 slice; only the third is genuinely E_1 in the sense that the Maurer–Cartan element cannot be recovered from the underlying E_∞-chiral algebra.

Tier (a): pole-free, R is the Koszul-signed flip. If $\mathcal{A}$ is pole-free (that is, the OPE $a(z)b(w)$ has no singularity for any $a, b \in \mathcal{A}$), then the degree-2 collision residue vanishes identically and $r(z) = 0$. In this tier the ordered bar complex admits a canonical Σ_n-equivariant structure: the descent datum R_σ on the local system $\mathcal{F}_n^{\mathrm{ord}}$ is the Koszul-signed permutation

$$R_\sigma \;=\; \varepsilon(\sigma)\cdot\tau_\sigma,$$

where τ_σ permutes tensor factors. The ordered and symmetric bar complexes are quasi-isomorphic in this tier. Every genuinely commutative (in the E_∞ sense) example lives here: the trivial chiral algebra, free-field tensor products with vanishing cross-OPE, and the structure sheaf of a point.

Tier (b): vertex algebras, R from local OPE. If $\mathcal{A}$ is a vertex algebra in the Borcherds–Frenkel–Lepowsky–Meurman sense (more generally, any E_∞-chiral algebra with poles in its OPE), then $r(z)$ is determined by the degree-2 residue of the OPE along the Arnold form $d\log(z_1 - z_2)$. All standard families of conformal field theory live here: Heisenberg $\mathcal{H}_k$ with $r(z) = k/z$, affine Kac–Moody $\widehat{\mathfrak{g}}_k$ with $r(z) = k\Omega_{\mathfrak{g}}/z$, Virasoro Vir_c with $r(z) = (c/2)/z^3 + 2T/z$ (cubic and simple poles, no quartic), $\beta\gamma$, bc, W-algebras, Bershadsky–Polyakov. In each case $r(z)$ is fully determined by the underlying E_∞ vertex algebra structure; the E_1 refinement adds no new data at degree 2 beyond what is already visible in the OPE. The averaging map on tier (b) recovers the scalar degree-2 shadow: for Heisenberg $\mathrm{av}(r(z)) = \kappa(\mathcal{A})$, while for non-abelian affine Kac–Moody $\mathrm{av}(r(z)) = \kappa_{\mathrm{dp}}(\mathcal{A})$ and the full $\kappa(\mathcal{A})$ adds the Sugawara shift $\dim(\mathfrak{g})/2$; the kernel is nontrivial but reconstructible from $\mathcal{A}$.

Tier (c): genuinely E_1. The third tier is populated by algebras in which $r(z)$ is independent input: the Maurer–Cartan element cannot be recovered from any underlying E_∞ structure. The paradigmatic examples are the Yangians $Y(\mathfrak{g})$ and the Etingof–Kazhdan quantum vertex algebras [4] obtained from quasi-triangular Lie bialgebras. In this tier the degree-2 Maurer–Cartan element is the rational Yang–Baxter r-matrix $r_{\mathrm{YB}}(z)$, and the associator is a full Drinfeld associator. Existing vertex-algebra theory cannot recognise these algebras, and the ordered bar complex is the minimal framework that makes them visible.

Remark 6.1 (E_1 is the atom of the nonlocal regime). The three tiers form a strict hierarchy. Every standard chiral algebra is E_∞ in the sense that its OPE is symmetric in the operator ordering up to the signs from the Arnold form; it is the interpretation as an E_1 algebra that is the refinement. Tier (c) consists of genuinely nonlocal objects, in the precise sense that the Maurer–Cartan element lies outside the image of any E_∞-to-E_1 promotion functor. The error of calling E_∞-chiral algebras "local" and E_1-chiral algebras "non-abelian" is a recurring pitfall; the right distinction is symmetric versus ordered, or closed versus open. The averaging map of Section 4 erases the ordering.

7. Four functors, four distinct objects

We close the structural part of the paper with a table of functors. Four distinct functors act on a cyclic E_1-chiral algebra $\mathcal{A}$, and each produces a different object. Care is required in keeping them apart.

(i) *Bar.* The ordered bar functor $\mathcal{A} \mapsto \overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$. The output is a coalgebra over the Feynman transform $F\mathrm{Ass}$ of the ribbon modular operad.

(ii) *Cobar.* The cobar functor $C \mapsto \Omega^{\mathrm{ch}}(C)$ on $F\mathrm{Ass}$-coalgebras. On the PBW-complete locus of Theorem 5.2, the composite $\Omega^{\mathrm{ch}} \circ \overline{B}^{\mathrm{ord}}$ is quasi-isomorphic to the identity; this is a statement about the bar–cobar adjunction returning $\mathcal{A}$ to itself, and is structurally distinct from the functor that produces the dual algebra.

(iii) *Verdier.* The Verdier-duality functor $\mathcal{A} \mapsto \mathcal{A}^!_\infty$, where $\mathcal{A}^!_\infty$ is the chain-level dual on the Ran space. Verdier duality intertwines the bar functor in the sense of [9, Theorem A]; the strict linear Koszul dual $\mathcal{A}^!$, obtained by taking a finite-dimensional model and applying linear duality, in general differs from $\mathcal{A}^!_\infty$ and the two must be kept distinct.

(iv) *Derived centre.* The chiral Hochschild cochain functor $\mathcal{A} \mapsto Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = \mathrm{HH}^{*,\mathrm{ch}}(\mathcal{A})$. In the open-closed picture this plays the role of the bulk: the centre of the open (line-operator) algebra $\mathcal{A}$ is the closed (bulk) sector. This functor is *not* the bar–cobar composition; it is a separate construction, the subject of Theorem H.

Theorem A of Volume I is the compatibility statement between (i) and (iii): the bar functor intertwines with Verdier duality on the Ran space. Theorem B (its E_1 version, Theorem 5.2 above) is the internal statement about (i) and (ii): on the PBW-complete locus the bar-cobar composite returns $\mathcal{A}$ up to quasi-isomorphism. Theorem H is the statement about (iv): the derived centre coincides with the ordered chiral Hochschild complex of the bar coalgebra. The four objects are four different outputs; the statement "$\overline{B}^{\mathrm{ord}}(\mathcal{A})$ computes the bulk" is category-theoretically ill-formed, because the bulk is the derived centre, not the bar.

In the three-pillar language of Volume I, the distinction is: bar-cobar adjointness gives inversion; Verdier duality gives $\mathcal{A}^!$; chiral Hochschild cochains give the bulk. The averaging map av of Section 4 intertwines the ordered and symmetric versions of the bar functor, and descends to the averaged version of each of the other three functors as well.

8. Conclusion and open problems

We have argued that the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, equipped with its deconcatenation coproduct, is the natural primitive object of modular Koszul duality, and that the symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ is its Σ_n-coinvariant shadow under an averaging map av of dg Lie algebras. The five foundational theorems A, B, C, D, H of modular Koszul duality admit ordered E_1 counterparts whose symmetric images are the classical theorems. Theorem D is the cleanest example: the scalar degree-2 shadow is recovered from the spectral r-matrix $r(z)$ by averaging, giving κ for abelian families and κ_{dp} for non-abelian affine Kac–Moody before the Sugawara shift $\dim(\mathfrak{g})/2$ is added; the passage from $r(z)$ to the scalar shadow is a strict projection with nontrivial kernel.

Several problems remain open.

The full E_1 Koszul duality theorem. An unconditional E_1 Koszul duality theorem, asserting that the bar and cobar functors exchange the categories of E_1-chiral algebras and cocomplete $F\mathrm{Ass}$-coalgebras on the Koszul locus, is not available in the present paper. What is proved (Theorems 5.1 and 5.2) is the existence of the $F\mathrm{Ass}$-coalgebra structure on the ordered bar, the inversion quasi-isomorphism on the Koszul locus, and the compatibility with averaging. A full equivalence of categories requires a tighter control on the coderived category of $F\mathrm{Ass}$-coalgebras and a characterisation of the Koszul locus that is independent of the commutative shadow.

The line-side modular R-matrix at higher genus. Theorem 5.4 packages the ordered degree-2 shadow into a formal series $R^{E_1,\mathrm{bin}}(z;\quad)$ whose g-th coefficient is the ordered degree-2 correction at genus g. Whether this formal series has a line-side interpretation as a genuine higher-genus modular R-matrix (that is, an R-matrix for a quantum group deformed over the moduli of curves) is a separate question. It would be expected on physical grounds from the 't Hooft expansion of ribbon graphs, and the cyclic trace formalism of the ribbonised Swiss-cheese construction provides the combinatorial language, but no proof exists in the cases beyond genus one.

GRT_1-torsor structure of splittings. The non-splitting proposition (Proposition 4.6) asserts that the space of dg Lie sections of av compatible with the Maurer–Cartan equation is a torsor for GRT_1. Writing down an explicit splitting (equivalently, choosing an associator) would give a canonical promotion of the symmetric theory to the ordered theory. The Etingof–Kazhdan theorem provides such splittings for Lie bialgebras; whether the analogous statement for chiral algebras produces a functorial lift of $\kappa \mapsto r(z)$ remains open.

The Swiss-cheese and curved sectors. The Swiss-cheese operad $\mathrm{SC}^{\mathrm{ch,top}}$, in its two-coloured open-closed form, is the natural habitat of the averaging map when one works with both sectors simultaneously. Promotion from a single-colour E_1-chiral algebra to the Swiss-cheese setting takes $\mathcal{A}$ to the pair $(\mathcal{A}, \mathcal{A})$ with closed sector $B_{\mathrm{Com}}(\mathcal{A})$ and open sector $B_{\mathrm{Ass}}(\mathcal{A})$ plus a mixed sector of dimension $(k-1)!\binom{k+m}{m}$ at Swiss-cheese degree (k, m). The averaging map restricted to the open sector is the subject of the present paper; the analogous statement in the mixed sector, which governs brane boundaries and open-closed coupling, is developed in Volume II but is not yet proved in the form stated here. A full Beilinson-audited Koszul theorem for the Swiss-cheese operad would give a unified statement absorbing both the closed and open sides.

We close by reiterating the thesis in one sentence: in modular Koszul duality, the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is the primitive object and the symmetric bar is its Σ_n-coinvariant shadow; the five theorems A, B, C, D, H are the invariants that survive the averaging, and Theorem D displays this most visibly at degree 2: for Heisenberg $\mathrm{av}(r(z)) = \kappa(\mathcal{A})$, while for non-abelian affine Kac–Moody $\mathrm{av}(r(z)) = \kappa_{\mathrm{dp}}(\mathcal{A})$ and the full $\kappa(\mathcal{A})$ adds the Sugawara shift $\dim(\mathfrak{g})/2$.

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Appendix A. Note added in proof: ordered Koszul chiral algebra, Yangian instance, chiral QG equivalence (ordered)

The 2026-04-16 closure wave inscribed three advances that directly upgrade the open-problem section (§8) of this note to partial theorems on the E_1-chiral side of the atlas.

Definition of ordered Koszul chiral algebra.

Definition A.1 (Ordered Koszul chiral algebra; N3-NIP)**.** A chiral algebra $\mathcal{A}$ on a smooth curve X is *ordered Koszul* if:

(a) the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct carries a deformation-retract onto its cohomology as a coalgebra over $(\mathrm{ChirAss})^!$;

(b) the counit $\Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ord}}(\mathcal{A})) \to \mathcal{A}$ is a quasi-isomorphism in the Francis–Gaitsgory factorization ambient;

(c) the R-twisted descent from ordered to symmetric bar (Vol I `lem:R-twisted-descent`, under Yang–Baxter plus unitarity $R(z)\,R^{op}(-z) = \mathrm{id}$; automatic for rational Yangians and generic-q trigonometric $U_q(\widehat{\mathfrak{g}})$, vacuous when $R(z) = \tau$) is compatible with the chiral twisting morphism.

This is the formal counterpart of `def:ordered-koszul-chiral-algebra` in Vol I `chapters/theory/chiral_qg_equiv_ord`

Yangian ordered Koszul (proposition).

Proposition A.2 ($Y(\mathfrak{g})^{\mathrm{ch}}$ is ordered Koszul)**.** *The chiral Yangian $Y(\mathfrak{g})^{\mathrm{ch}}$, defined via the Drinfeld second presentation and Etingof–Kazhdan quantisation flatness, is an ordered Koszul chiral algebra in the sense of Definition* A.1*.*

Proof sketch. By Vol I `prop:yangian-ordered-koszul` (`chapters/theory/chiral_qg_equiv_ordered.tex`). The Drinfeld second-presentation generators assemble an ordered bar complex whose cohomology is computed by pure-braid Orlik–Solomon (Shelton–Yuzvinsky Koszulity), and Etingof–Kazhdan flatness supplies the chiral twisting morphism. □

This upgrades the open problem "The full E_1 Koszul duality theorem" of §8 to: *Koszul duality holds on the ordered Koszul locus, and the Yangian is a concrete instance.*

Chiral QG equivalence on the ordered Koszul locus.

Theorem A.3 (Chiral quantum-group equivalence, ordered Koszul locus)**.** *On the ordered Koszul locus (Definition* A.1*), three structures on a chiral algebra $\mathcal{A}$ determine each other up to Drinfeld associator:*

(i) *the classical R-matrix $r_{\mathcal{A}}(z)$;*

(ii) *the A_∞-structure on $\overline{B}^{\mathrm{ord}}(\mathcal{A})$;*

(iii) *the deconcatenation coproduct on $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ with compatible chiral twisting morphism.*

Cohomologically-derived Center is associator-INDEPENDENT (verified for $\mathfrak{sl}_2$ at order); cochain-level data is associator-DEPENDENT with the dependence classified by GRT_1.

Proof sketch. Vol I `thm:chiral-qg-equiv-ordered` (`chapters/theory/chiral_qg_equiv_ordered.tex`), with explicit $\mathfrak{sl}_2$ Yangian triangle (`prop:sl2-yangian-triangle-concrete`) demonstrating triangle coherence $\alpha \circ \gamma \circ \beta = \mathrm{id}$ at order . □

Two further ordered-locus theorems. (N3-NIP.a) gl_N **Drinfeld-double internal.** Vol I `thm:glN-drinfeld-double-internal` (`chapters/theory/chiral_qg_equiv_ordered.tex`) bypasses the JKL26 phantom dependency: the gl_N chiral Drinfeld double is constructed internally via Miura + cross-arity Stasheff + $U(\hat{\mathfrak{gl}}_1)$ screenings. Two boundary parameters: degenerate $\Psi = 0$ recovers the free abelian chiral algebra (identity quantum group); $\Psi = 1$ is the free-boson point where the OPE obstruction vanishes.

(N3-NIP.b) $\mathcal{W}_\infty$ chiral QG weight-completed. `thm:w-infty-chiral-qg-completed` resolves the class-`M` contradiction through the weight-completed category, in parallel with the MC5 weight-completed route.

(N3-NIP.c) GRT_1**-rigidity of cohomological center.** `thm:grt1-rigidity` makes associator-dependence explicit: *trivial on H^0; homotopic on cochains*. This sharpens Proposition 4.6 of the present note: the GRT_1-torsor of splittings acts trivially on the derived center at the H^0 level but non-trivially at the cochain level.

Elliptic and toroidal extensions. (N3-NIP.d) Elliptic. `thm:chiral-qg-equiv-elliptic` covers the genus-1 case with Felder dynamical $R(z,\tau)$ and Fay trisecant replacing the pentagon identity. Fay trisecant extends to all $g \geq 0$ universally; the distinct theta-nulls quartic identity on the theta divisor is a separate statement at $g \geq 2$.

(N3-NIP.e) Toroidal formal disk. `thm:chiral-qg-equiv-toroidal-sf` of `standalone/seven_faces.tex` (inscribed; not an `\input` of `main.tex`, hence a standalone inscription whose canonical formal-disk home is Vol III `quantum_groups_foundations.tex`) gives, via Schiffmann–Vasserot CoHA plus Ding–Iohara–Miki presentation of $U_{q,t}(\ddot{\mathfrak{sl}}_N)$ with two spectral parameters (u,v), chiral QG equivalence at the formal-disk level. The earlier `thm:chiral-qg-equiv-toroidal-formal-disk` label was never inscribed as a `\label` anywhere in Vol I; the canonical standalone label is the `-sf` form above. The global $\mathbb{P}^1 \times \mathbb{P}^1$ case is conditional on the class-`M` chain-level topologization frontier on the original bar complex in $\mathrm{Ch}(\mathrm{Vect})$.

Averaging and the Sugawara shift: precise statement. For clarity we repeat the Sugawara-shift correction at degree 2 (already present in the main text, here aggregated for reference):

- *Heisenberg $\mathcal{H}_k$:* $\mathrm{av}(r(z)) = \kappa(\mathcal{H}_k) = k$ (no shift; the r-matrix is scalar).
- *Non-abelian affine Kac–Moody $V_k(\mathfrak{g})$:* $\mathrm{av}(r(z)) = \kappa_{\mathrm{dp}}(V_k(\mathfrak{g})) = k\dim(\mathfrak{g})/(2h^\vee)$; the full $\kappa(V_k(\mathfrak{g})) = \mathrm{av}(r(z)) + \dim(\mathfrak{g})/2 = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$ adds the Sugawara shift $\dim(\mathfrak{g})/2$.
- *Virasoro Vir_c:* $r(z) = (c/2)/z^3 + 2T/z$ (cubic and simple poles); averaging gives $\kappa(\mathrm{Vir}_c) = c/2$ without a further shift.
- *Principal $\mathcal{W}_N$:* $\kappa(\mathcal{W}_N) = c \cdot (H_N - 1)$ with $H_N = \sum_{j=1}^N 1/j$ (NOT H_{N-1}; at $N = 2$, $H_2 - 1 = 1/2$ recovers $\kappa(\mathrm{Vir}_c) = c/2$).

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